

APPENDIX:

Socioeconomic Segregation and Park Greenness: Insights Across a Strong Latitudinal Gradient

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1. Appendix

1.1. Spatial Characterization of Study Areas

To provide a comprehensive spatial context for each analyzed conurbation, we present detailed maps depicting the four core variables integrated in this study. These maps illustrate the spatial correspondence between public parks, neighborhood socioeconomic status, vegetation greenness (NDVI), and social mixing patterns. Visual comparison of these layers complements the quantitative analyses presented in the main text by highlighting the spatial overlap between high-SES neighborhoods, greener parks, and higher levels of social mixing. Complete map sets for all study conurbations are available as supplementary materials in the project's GitHub repository.

1.2. Analysis of Variance

The ANOVA helps to determine the statistical differences between the average greenness associated to socioeconomic quintiles of green urban areas.

Table 1: One-way ANOVA results testing differences in mean park greenness (NDVI) among neighborhood socioeconomic status (ISMT) quintiles for each study city. Reported statistics include degrees of freedom (DF), sums of squares (Sum Sq), mean squares (Mean Sq), F-values, and associated p-values.

	A	B	C	D	E	F
	Antofagasta	La Serena	Valparaíso	Santiago	Concepción	Puerto Montt
DF						
Q	2	3	3	3	3	3
Residuals	217	998	783	9808	1596	621
Sum Sq						
Q	0.129	0.070	0.198	17.370	2.507	0.366
Residuals	0.420	2.757	5.142	46.317	16.025	0.014
Mean Sq						
Q	0.065	0.023	0.066	5.79	0.836	0.122
Residuals	0.002	0.003	0.007	0.005	0.010	0.014
F Value	33.433	8.410	10.042	1226	83.224	8.598
p-value	2×10^{-13}	10^{-5}	7×10^{-5}	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	$< 2 \times 10^{-5}$

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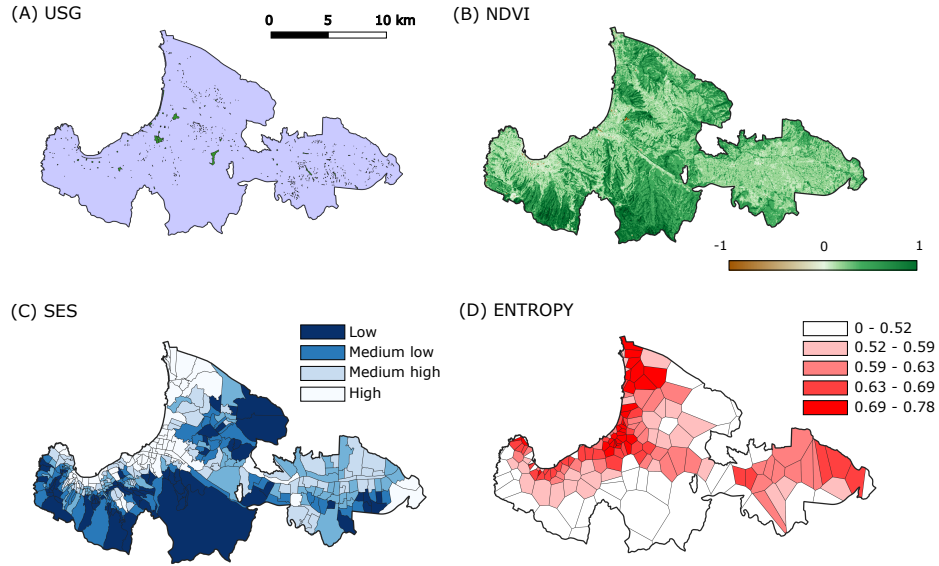


Fig. 1: Spatial distribution of key urban variables in Valparaíso. The panels display the four primary spatial datasets used in the analysis: **(A)** UGS. Location and extent of public urban green spaces (Urban Green Spaces). **(B)** NDVI. Greenness measured by the Normalized Difference Vegetation Index, where darker green indicates higher density. **(C)** Entropy - Voronoi. Social mixing levels calculated for each mobile phone antenna's Voronoi cell, with red indicating higher entropy (less segregation). **(D)** SES. Socioeconomic status distribution based on the Socio-Material Territorial Index (ISMT), where darker blue represents higher socioeconomic levels.

Tukey tests

Tukey tests determines the specific quintiles exhibiting statistical significant differences of average greenness between green urban areas stratified by its socioeconomic quintile.

Dendrograms

We here show dendrograms for the five remaining cities, *i.e.* La Serena, Valparaíso, Santiago, Concepción and Puerto Montt.

Table 2: Tukey post-hoc analysis shows the mean NDVI differences between ISMT quartiles. Significant comparisons between groups I and J are highlighted and marked by an asterisk (*i.e.* $p < 0.05$).

City	(I) ISMT Quartile	(J) ISMT Quartile	Mean difference (I - J) Δ NDVI	95% Confidence Interval (lower, upper)
A - Antofagasta	Medium Low	Medium High	0.007	(-0.015, 0.029)
		High	-0.045*	(-0.064, -0.025)
	Medium High	High	-0.052*	(-0.068, -0.035)
B - La Serena	Low	Medium Low	0.003	(-0.013, 0.018)
		Medium High	-0.019*	(-0.032, -0.007)
		High	-0.011	(-0.024, 0.001)
	Medium Low	Medium High	-0.022*	(-0.036, -0.008)
		High	-0.014*	(-0.027, -0.001)
C - Valparaíso	Low	Medium Low	0.008	(-0.002, 0.018)
		Medium High	0.015	(-0.006, 0.037)
		High	-0.031*	(-0.055, -0.007)
	Medium Low	Medium High	-0.022*	(-0.041, -0.003)
		High	-0.046*	(-0.068, -0.024)
D - Santiago	Low	High	-0.025*	(-0.047, -0.002)
		Medium Low	-0.020*	(-0.024, -0.015)
		Medium High	-0.040*	(-0.045, -0.035)
	Medium Low	High	-0.125*	(-0.131, -0.120)
		Medium High	-0.020*	(-0.026, -0.015)
E - Concepción	Low	High	-0.106*	(-0.111, -0.100)
		High	-0.085*	(-0.091, -0.079)
		Medium Low	-0.023*	(-0.044, -0.001)
	Medium Low	Medium High	-0.053*	(-0.070, -0.037)
		High	-0.101*	(-0.118, -0.084)
F - Puerto Montt	Low	Medium High	-0.031*	(-0.054, -0.008)
		High	-0.079*	(-0.102, -0.056)
		High	-0.048*	(-0.066, -0.030)
	Medium Low	Medium Low	-0.057*	(-0.110, -0.004)
		Medium High	-0.002	(-0.039, 0.036)
	Medium Low	High	-0.049*	(-0.088, -0.010)
		Medium High	0.055*	(0.009, 0.100)
	Medium High	High	0.008	(-0.039, 0.055)
	Medium High	High	-0.047*	(-0.075, -0.019)

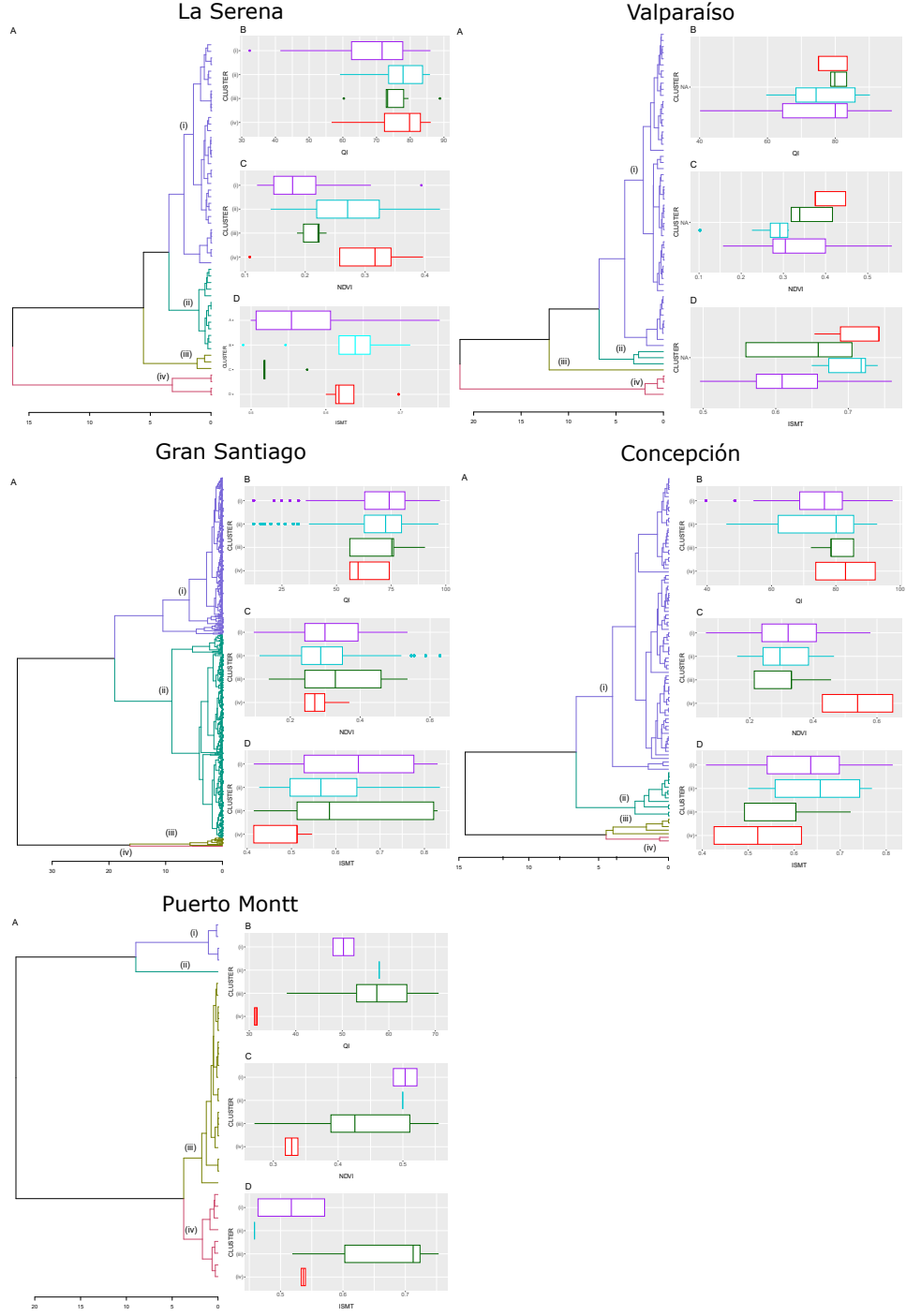


Fig. 2: Cluster analysis of the entropy timelines of public parks for La Serena, Valparaíso, Snatiago, Concepción and Puerto Montt. **A** shows the dendrogram arbitrarily trimmed to four clusters, *i.e.* (i) to (iv) represent parks sharing a similar patterns of entropy timelines across the day. Values on the *x-axis* show the distance between clusters evaluated using the UPMGA algorithm. **B** shows the boxplot of the Quality Index of public parks among clusters in **A**. **C** shows the differences between mean NDVI values across clusters. **D** the mean ISMT distributed across clusters. note that Antofagasta is shown in the main text of the manuscript.